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Abstract

I will describe the main project that can be carried out during the School and that students can potentially continue or develop further after the end of the School. The project focuses on the identification and characterization of merger debris via clustering algorithms using Gaia data.

1 Identification and characterization of merger debris through clustering using Gaia data

The goal of the project is to reproduce and then extend the analysis carried out in Dodd et al. [2023]. This study constructed a local sample of halo stars (defined kinematically) from Gaia DR3 data, run the clustering algorithm developed by Lövdal et al. [2022], and characterized the debris [see Ruiz-Lara et al., 2022, Ruiz-Lara et al., 2022, Dodd et al., 2024, Matsuno et al., 2022, Horta et al., 2023, Dodd et al., 2025]. For general reviews on the topic please consult Helmi [2020], Deason and Belokurov [2024].

1.1 Dataset

The needed dataset will include 6D phase-space coordinates and magnitudes G, G_{BP}, G_{RP} information. There are two possibilities:

1. The student gets an account on the Gaia archive (`https://gea.esac.esa.int/archive/`) or Datalabs (recommended for the future) `https://datalabs.esa.int/`and constructs their own sample following the steps in Dodd et al. [2023]

2. We place the data in a publicly available location (`gaia-dr3-rv-poege5.hdf5`, 34Gb). The data can be downloaded from

   `https://filesender.surf.nl/?s=download&token=2fde27c7-3bf2-48f5-ab34-ae7bb3498ea2`

1.2 Software

There are many clustering algorithms that could be used, but Lövdal et al. [2022] data-driven algorithm is the preferred choice.

See https://github.com/SofieLovdal/IOM_clustering

Then the package to identify membership of stars can be obtained from https://github.com/TomCallingham/id$_$Dodd23/blob/main/README.md and instructions therein. This package has utilities to:

- Transform from observables to the Galactocentric position-velocity used in Dodd23
- Calculate IoM used in Dodd et al. [2023] (Energy, Lz, L perp)
- Identify the membership of the stars to substructures in Dodd et al. [2023].

See the examples notebook. Once the members have been identified proceed to construct colour-magnitude diagrams, metallicity distributions, C=compute and compare the mass functions [Offner, 2026, Li et al., 2026], etc.

1.3 Beyond this project

Possible extensions include:

- Extend the analysis to beyond the Solar neighbourhood (it requires revisiting the selection of halo stars, see e.g. [Posti et al., 2018, Veljanoski et al., 2019])
- Complementing Gaia data with other surveys for radial velocities and possibly also for metallicities; notably DESI has provided a recent data release [see, e.g. Ruiz-Lara et al., 2022, Koposov et al., 2026].
- By integrating the orbits forwards and backwards in time of known members of a given substructure, search for additional members using Gaia DR3, and shed light on the nature of the structure using their CMD or metallicity distribution function [see e.g. Lahoye, 2025, Nogueira-Santos et al., 2026].
- Use Gaia DR4 when released

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